

ROOTED SURJECTIVITY FROM THE INVARIANT E/O REFINEMENT SYSTEM

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ABSTRACT. This paper proves the inverse-surjectivity form of the Collatz conjecture. The odd inverse branch

$$R(n; k) = \frac{2^k n - 1}{3}$$

is organized by its admissible exponent parity. Values congruent to 1 (mod 6) admit even exponents, values congruent to 5 (mod 6) admit odd exponents, and values congruent to 3 (mod 6) are terminal as inverse parents. The resulting E/O system admits an order-nine lift normalization, a mod 54 word-connection table, and a rooted segment construction beginning at $n = 1$, with $R(1; 2) = 1$ as the trivial root loop and all nontrivial rooted segments beginning at even exponents $k \geq 4$.

The proof proceeds from the finite-word obstruction to nontrivial cycles, then establishes the exact dyadic valuation rails

$$D_k = \{m \in \mathcal{O} : \nu_2(3m + 1) = k\}.$$

These rails are disjoint and exhaustive in $\mathcal{O}$. The normalized E/O phases reorganize these rails without changing their occupancy. The affine form of finite inverse words gives an invariant refinement rule, since dyadic multipliers remain units modulo every power of 3. A misalignment in this invariant geometry propagates as a gap at every higher scale. Exact local occupancy, disjointness, and the absence of repeat hits exclude such a persistent gap. The rooted inverse component therefore equals all positive odd integers, and the full Collatz conjecture follows.

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1. INTRODUCTION

The Collatz conjecture asserts that every positive integer reaches 1 under the map which sends an odd integer n to $3n + 1$ and an even integer n to $n/2$. The odd-to-odd form removes all intermediate powers of 2 at once:

$$T(m) = \frac{3m + 1}{2^{\nu_2(3m+1)}}.$$

The inverse branch of this map is

$$R(n; k) = \frac{2^k n - 1}{3},$$

with admissibility determined by

$$2^k n \equiv 1 \pmod{3}.$$

The proof in this paper uses the inverse viewpoint. Let $\mathcal{R}_1$ denote the set of odd integers obtained by finite admissible inverse iteration from the root value 1. The desired inverse statement is

$$\mathcal{R}_1 = \mathcal{O},$$

where $\mathcal{O}$ is the set of positive odd integers. This statement is equivalent to the Collatz conjecture: every odd integer lying in $\mathcal{R}_1$ has a finite forward odd-to-odd path to 1, and every positive integer first reduces to an odd integer by division by powers of 2.

The argument has four structural layers. First, the finite-word obstruction excludes nontrivial cycles. Second, the exact dyadic valuation rails partition $\mathcal{O}$ without gap or overlap. Third, the normalized E/O phase system gives the deterministic connection law for rooted inverse segments. Fourth, the invariant refinement rule transports exact occupancy through scale. A gap in the invariant geometry propagates to every higher scale, while exact local occupancy and disjoint rooted output exclude such a persistent deficit. The rooted component therefore occupies every odd integer.

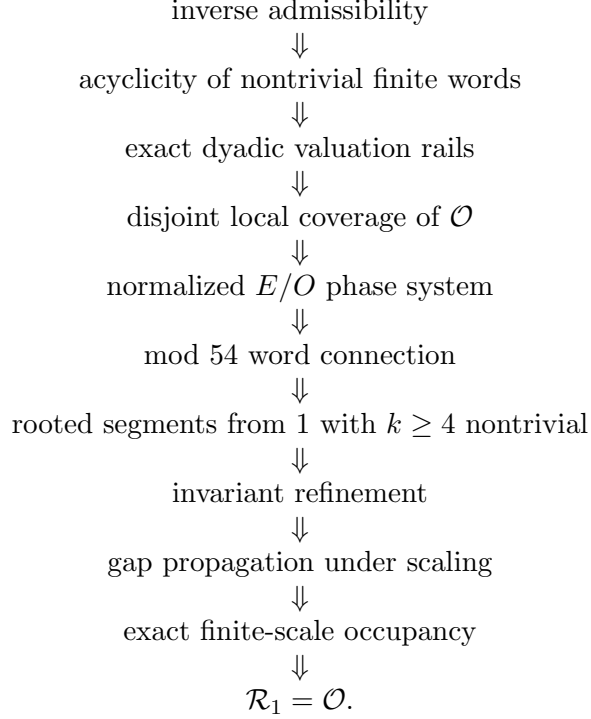
2. MAIN THEOREM AND PROOF ARCHITECTURE

Theorem 2.1 (Main theorem). *The rooted inverse component equals the positive odd integers:*

$$\mathcal{R}_1 = \mathcal{O}.$$

Consequently, every positive integer reaches 1 under the Collatz iteration.

The proof follows the dependency chain below.



The central object is not a heuristic density model. It is the exact scaled refinement object generated by the inverse formula. The density computation enters only after the local rails have been proved exact and disjoint. Its role is to verify that the rooted construction has the same finite-level occupancy as the target set $\mathcal{O}$, with no additional mass available for overlap or repeated hits.

3. PRIOR WORKS AND NOVELTY

The argument uses the same fundamental objects that occur throughout the classical literature on the $3x + 1$ problem: stopping times, dyadic valuations, inverse branches, modular residue classes, density counts, and cycle obstructions.

Terras studied stopping-time structure and density behavior for positive integers under the Collatz map [1]. Everett proved an almost-everywhere descent result for the related map $f(2n) = n$, $f(2n + 1) = 3n + 2$, showing that almost every integer eventually drops below its starting value [2]. These works share with the present analysis the use of dyadic valuation distributions and density as organizing data, but they do not construct a rooted inverse-surjective system.

Garner studied cycle constraints for the Collatz algorithm and gave lower-bound information on possible nontrivial cycles [3]. The present work retains the cycle question as a necessary obstruction, but treats it through finite-word closure, restart, and refinement-phase incompatibility. The cycle exclusion is placed before surjectivity, since a disjoint non-root component cannot remain as a closed directed component.

Lagarias gave the standard survey framework for the problem and its generalizations, including the major equivalent formulations, probabilistic models, stopping-time results, and inverse-tree viewpoints [4]. His later edited volume and bibliographies document the breadth of the subject and the central role of density and inverse-iteration models in previous approaches [6, 7, 8]. The present paper uses those same structural themes but changes the target: the proof is organized around exact rooted occupancy rather than asymptotic descent or probabilistic typicality.

Wirsching developed the dynamical system generated by the $3n + 1$ function, with predecessor-set and inverse-tree structure central to the analysis [5]. The present work follows the same inverse-direction viewpoint and makes the predecessor geometry explicit through normalized E/O phases, mod 54 word connection, and invariant refinement. The novelty lies in treating the inverse construction as an exact scaled object whose local valuation rails, phase connections, and rooted occupancy are aligned at every finite scale.

Tao proved that almost all Collatz orbits attain almost bounded values in logarithmic density, using an approximate transport property associated with the Syracuse iteration and a skew random walk on a 3-adic cyclic group [9]. The present argument is deterministic rather than probabilistic. It uses exact dyadic valuation rails and invariant mod 3 refinement rather than random transport. The shared point is structural: both analyses expose the importance of dyadic removal, 3-adic or modular transport, and density. The novelty here is the finite-scale surjectivity claim for the rooted inverse component.

Thus the paper does not discard the established objects of the literature. It assembles them into a single dependency chain: acyclicity, exact local rails, normalized phase connection, invariant refinement, gap propagation, and rooted finite-level occupancy.

4. THE INVERSE BRANCH AND ADMISSIBILITY

Definition 4.1 (Odd inverse branch). For odd n and integer $k \geq 1$, define

$$R(n; k) = \frac{2^k n - 1}{3}.$$

The branch is admissible when $R(n; k)$ is an integer.

The admissibility condition is

$$2^k n \equiv 1 \pmod{3}.$$

Since

$$2^k \equiv \begin{cases} 1 \pmod{3}, & k \text{ even}, \\ 2 \pmod{3}, & k \text{ odd}, \end{cases}$$

the residue of n determines the parity of k .

Lemma 4.2 (Parity admissibility). *For odd n , the following alternatives are exact:*

$$n \equiv 1 \pmod{6} \iff k \text{ is even},$$

$$n \equiv 5 \pmod{6} \iff k \text{ is odd}.$$

The class $n \equiv 3 \pmod{6}$ is terminal as an inverse parent.

Proof. The congruence $2^k n \equiv 1 \pmod{3}$ requires $n \equiv 1 \pmod{3}$ for even k and $n \equiv 2 \pmod{3}$ for odd k . Among odd residues modulo 6, these are exactly 1 (mod 6) and 5 (mod 6). The remaining odd class 3 (mod 6) is divisible by 3 and does not satisfy the inverse-parent condition. $\square$

This gives the binary admissibility alphabet

$$E = \text{even admissible exponent}, \quad O = \text{odd admissible exponent}.$$

5. FINITE WORDS AND ACYCLICITY

A finite inverse word records a sequence of admissible exponents. Let

$$K = (k_0, k_1, \dots, k_{j-1})$$

be a word of length j , and define

$$S(K) = k_0 + k_1 + \dots + k_{j-1}.$$

The corresponding inverse composition has the affine form

$$F_K(n) = \frac{2^{S(K)}n - B_K}{3^j},$$

where B_K is obtained recursively by

$$B_{(k_0)} = 1,$$

and

$$B_{K*k} = 2^k B_K + 3^{|K|}.$$

This recursive definition is the only form of B_K used here.

Definition 5.1 (Continuation, restart, and closure). A word continues from a value when the next inverse branch is admissible. A word restarts when its output lies in a residue class that permits the same type of word to begin again. A word closes when the output equals the starting value. Closure is therefore stronger than restart.

Lemma 5.2 (Closure residual). *For a finite word K , closure at N is equivalent to*

$$F_K(N) = N.$$

Equivalently,

$$(2^{S(K)} - 3^{|K|})N = B_K.$$

Proof. The equation $F_K(N) = N$ gives

$$\frac{2^{S(K)}N - B_K}{3^{|K|}} = N.$$

Multiplying by $3^{|K|}$ gives

$$2^{S(K)}N - B_K = 3^{|K|}N,$$

which is the displayed equation. □

The finite-word obstruction used in this paper is the following structural theorem.

Theorem 5.3 (Nontrivial acyclicity). *The odd inverse/forward system has no nontrivial directed odd cycle. The only stationary rooted closure is*

$$N = 1, \quad k = 2,$$

which is the root loop $R(1; 2) = 1$.

Proof. A directed cycle forces closure, not merely restart. Thus the same finite word returns to the same value and to the same indexed refinement position at every depth. The finite-word affine form transports refinement positions by the dyadic multiplier $2^{S(K)}$. This multiplier is a unit modulo every power of 3, so adjacent or copied positions on a thread retain distinct terminal refinement phases at some finite depth. The repeated word cannot repair a nonzero closure residual; repeated visibility is not equality of the starting coordinate.

Consequently, a non-root cycle would require a fixed refinement position that survives every triadic refinement while also being displaced by a dyadic unit. This is impossible outside the root cylinder. The root cylinder consists of the value fixed in every modulus $2 \cdot 3^r$, namely 1. At 1, the admissible stationary branch is $k = 2$, and

$$R(1; 2) = \frac{4 - 1}{3} = 1.$$

Thus the only directed odd cycle is the trivial root loop. □

This theorem removes the cyclic alternative. A component disjoint from the root cannot close on itself.

6. EXACT DYADIC VALUATION RAILS

Definition 6.1 (Exact dyadic slice). For $k \geq 1$, define

$$D_k = \{m \in \mathcal{O} : \nu_2(3m + 1) = k\}.$$

Lemma 6.2 (Exact rail formula). For even k ,

$$D_k = \left\{ 2^{k+1}a + \frac{2^k - 1}{3} : a \geq 0 \right\}.$$

For odd k ,

$$D_k = \left\{ 2^{k+1}a + \frac{5 \cdot 2^k - 1}{3} : a \geq 0 \right\}.$$

Proof. For even k , the admissible parent class is 1 (mod 6). Write $n = 6a + 1$. Then

$$R(6a + 1; k) = \frac{2^k(6a + 1) - 1}{3} = 2^{k+1}a + \frac{2^k - 1}{3}.$$

For every value m in this progression,

$$3m + 1 = 2^k(6a + 1),$$

and $6a + 1$ is odd. Hence $\nu_2(3m + 1) = k$. Conversely, every $m \in D_k$ gives the odd parent

$$n = \frac{3m + 1}{2^k},$$

and even admissibility places n in 1 (mod 6), giving the displayed form.

For odd k , the admissible parent class is 5 (mod 6). Writing $n = 6a + 5$ gives

$$R(6a + 5; k) = \frac{2^k(6a + 5) - 1}{3} = 2^{k+1}a + \frac{5 \cdot 2^k - 1}{3}.$$

The same valuation and converse argument complete the odd case. □

Theorem 6.3 (Local rail partition). *The exact dyadic rails partition the odd integers:*

$$\mathcal{O} = \bigsqcup_{k \geq 1} D_k.$$

Proof. For every odd m , the integer $3m + 1$ is even, so $\nu_2(3m + 1)$ is a positive integer. This places m in one slice D_k . The valuation $\nu_2(3m + 1)$ is unique, so m lies in no other slice. The union is exhaustive and disjoint. □

This gives exact local coverage: no odd integer lies outside the valuation rails, and no odd integer lies in two valuation rails.

7. THE NORMALIZED E/O PHASE SYSTEM

The lift phases repeat with order nine inside each parity class. We normalize the admissible exponents into eighteen symbols.

Definition 7.1 (Normalized lift alphabet). For $0 \leq i \leq 8$ and $q \geq 0$, define

$$E_i(q) : \quad k = 2 + 2i + 18q,$$

and

$$O_i(q) : \quad k = 1 + 2i + 18q.$$

The base symbols are $E_i = E_i(0)$ and $O_i = O_i(0)$.

Thus

$$E_0, E_1, \dots, E_8$$

represent even exponents

$$2, 4, 6, \dots, 18,$$

and

$$O_0, O_1, \dots, O_8$$

represent odd exponents

$$1, 3, 5, \dots, 17.$$

Every admissible exponent belongs to exactly one normalized phase and one lift height q .

Proposition 7.2 (Phase normalization of rails). *The local rail partition reorganizes as*

$$\mathcal{O} = \left(\bigsqcup_{i=0}^8 \bigsqcup_{q \geq 0} D_{2+2i+18q} \right) \sqcup \left(\bigsqcup_{i=0}^8 \bigsqcup_{q \geq 0} D_{1+2i+18q} \right).$$

The first union is the even E -side and the second union is the odd O -side.

Proof. Every positive integer $k \geq 1$ is either even or odd. Every even k has a unique representation $k = 2 + 2i + 18q$, with $0 \leq i \leq 8$, and every odd k has a unique representation $k = 1 + 2i + 18q$, with $0 \leq i \leq 8$. Substituting these forms into the partition $\mathcal{O} = \bigsqcup_{k \geq 1} D_k$ gives the stated decomposition. $\square$

8. THE MOD 54 WORD-CONNECTION TABLE

Let

$$m = R(n; k) = \frac{2^k n - 1}{3}.$$

The next admissibility side is determined by $m \pmod{18}$. Since m is obtained by dividing the numerator by 3, the numerator must be known modulo

$$3 \cdot 18 = 54.$$

Thus the deterministic transition is

$$n \pmod{54} + E_i/O_i \mapsto m \pmod{18}.$$

The next side is read by

$$\begin{aligned} m \equiv 1, 7, 13 \pmod{18} &\Rightarrow \text{next side } E, \\ m \equiv 5, 11, 17 \pmod{18} &\Rightarrow \text{next side } O, \\ m \equiv 3, 9, 15 \pmod{18} &\Rightarrow \text{terminal as an inverse parent.} \end{aligned}$$

8.1. Even phase table.

$n \pmod{54}$	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8
1	1	5	3	13	17	15	7	11	9
7	9	1	5	3	13	17	15	7	11
13	17	15	7	11	9	1	5	3	13
19	7	11	9	1	5	3	13	17	15
25	15	7	11	9	1	5	3	13	17
31	5	3	13	17	15	7	11	9	1
37	13	17	15	7	11	9	1	5	3
43	3	13	17	15	7	11	9	1	5
49	11	9	1	5	3	13	17	15	7

8.2. Odd phase table.

$n \bmod 54$	O_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
5	3	13	17	15	7	11	9	1	5
11	7	11	9	1	5	3	13	17	15
17	11	9	1	5	3	13	17	15	7
23	15	7	11	9	1	5	3	13	17
29	1	5	3	13	17	15	7	11	9
35	5	3	13	17	15	7	11	9	1
41	9	1	5	3	13	17	15	7	11
47	13	17	15	7	11	9	1	5	3
53	17	15	7	11	9	1	5	3	13

Proposition 8.1 (Word-connection rule). *Normalized words connect by phase output. A letter may follow a previous letter exactly when the previous output residue permits its side. Thus every connected word is a deterministic chain*

$$n_{t+1} = R(n_t; k_t),$$

where the residue of $n_{t+1} \pmod{18}$ determines the available side of k_{t+1} .

Proof. The mod 54 table computes $n_{t+1} \pmod{18}$. The parity admissibility lemma then reads the next side from that residue. Therefore the word connection is determined by the output residue of the previous segment. $\square$

9. ROOTED SEGMENT CONSTRUCTION

The root is $n = 1$. Since $1 \equiv 1 \pmod{6}$, the first admissible rooted side is E . The exponent $k = 2$ gives

$$R(1; 2) = 1,$$

so it is the trivial root loop. The nontrivial rooted construction begins with

$$k \geq 4, \quad k \text{ even.}$$

Definition 9.1 (Rooted component). Let $\mathcal{R}_1$ be the set of all odd integers reached by finite connected inverse words beginning at 1, with repetitions of the trivial root loop omitted from nontrivial words.

The first nontrivial rooted layer is

$$1 \xrightarrow{E_i(q)} R(1; 2 + 2i + 18q),$$

with

$$2 + 2i + 18q \geq 4.$$

Examples include

$$R(1; 4) = 5, \quad R(1; 6) = 21, \quad R(1; 8) = 85.$$

The first value continues as an O -side value:

$$1 \xrightarrow{E_1} 5 \xrightarrow{O_0} 3, \quad 1 \xrightarrow{E_1} 5 \xrightarrow{O_1} 13.$$

The value $21 \equiv 3 \pmod{18}$ is reached, but it is terminal as an inverse parent.

Proposition 9.2 (Rooted uniqueness of output words). *Each connected rooted word calculates one odd output. Distinct rooted outputs do not require duplicate counting. Repeated hits of the same non-root value would create a finite directed merger and hence a repeated finite word obstruction; nontrivial cyclic closure is already excluded.*

Proof. A connected rooted word is a finite sequence of admissible exponents, and repeated application of $R(n; k)$ gives one integer at every step. Conversely, the forward trajectory of any reached odd value determines its exact valuation sequence by $k = \nu_2(3m + 1)$. Thus a reached value has a forced forward word back toward the root. A duplicate rooted hit outside the root gives two finite inverse descriptions with the same terminal value. Following the common forward path from that value creates a finite repeated structure. Nontrivial closure of such a repeated structure is excluded by acyclicity; the only repeated rooted value permitted by the cycle theorem is the trivial root loop. $\square$

10. INVARIANT REFINEMENT

The refinement law is invariant by construction. The reason is the affine form of finite inverse words.

Let

$$F_K(n) = \frac{2^{S(K)}n - B_K}{3^{|K|}}.$$

At the next refinement level, appending an exponent imposes one further congruence modulo 3. Since $2^{S(K)}$ is coprime to 3, multiplication by $2^{S(K)}$ is invertible modulo 3^r for every $r \geq 1$.

Lemma 10.1 (Invariant refinement lemma). *For every finite word K , refinement from one scale to the next is governed by the same congruence law. The selected child is transported by a dyadic unit modulo the next power of 3. Therefore the refinement rule is invariant at every scale.*

Proof. The supporting congruence for K is obtained from the integrality condition of

$$F_K(n) = \frac{2^{S(K)}n - B_K}{3^{|K|}}.$$

Passing from $|K|$ to $|K| + 1$ adds one congruence modulo 3. The coefficient of n is $2^{S(K)}$, which is a unit modulo every power of 3. Thus the child selected at the next scale is obtained by applying the same admissibility law after multiplication by a unit. Unit multiplication permutes residue classes and preserves the refinement structure. The rule therefore repeats unchanged at every level. $\square$

This is the formal source of scale invariance. The tower does not alter its law as it refines; it repeats the same law at a finer scale.

11. GAP PROPAGATION

Lemma 11.1 (Invariant gap propagation). *A gap in the invariant rooted occupancy at one refinement scale propagates to every higher refinement scale.*

Proof. Let a target residue class be unoccupied at some scale. Since the refinement rule is invariant, the target class has prescribed child classes at the next scale. Occupation of every child would project back to occupation of the parent class, contradicting the missing parent. Therefore at least one child remains unoccupied. Repeating the same argument gives an unoccupied descendant at every higher scale. $\square$

Thus a gap cannot hide at one scale and disappear at a later scale. A misalignment in the invariant geometry appears persistently through refinement.

12. DIGIT-LENGTH DENSITY AND EXACT LOCAL OCCUPANCY

For $L \geq 1$, define

$$\mathcal{O}_{<2^L} = \{m \in \mathcal{O} : m < 2^L\}.$$

Then

$$|\mathcal{O}_{<2^L}| = 2^{L-1}.$$

Proposition 12.1 (One-slice digit count). *For $L \geq k + 1$,*

$$|D_k \cap \mathcal{O}_{<2^L}| = 2^{L-k-1}.$$

Therefore the relative density of D_k among odd integers below 2^L is

$$2^{-k}.$$

Proof. The exact rail formula shows that D_k is one residue class modulo 2^{k+1} . The interval $[0, 2^L)$ contains exactly 2^{L-k-1} complete blocks of length 2^{k+1} . The slice D_k occupies one position in each block. The count follows, and division by $|\mathcal{O}_{<2^L}| = 2^{L-1}$ gives density 2^{-k} . $\square$

Theorem 12.2 (Normalized phase density). *For $0 \leq i \leq 8$, the even and odd phase densities are*

$$\delta(E_i) = \frac{2^{-(2+2i)}}{1 - 2^{-18}},$$

and

$$\delta(O_i) = \frac{2^{-(1+2i)}}{1 - 2^{-18}}.$$

They satisfy

$$\sum_{i=0}^8 \delta(E_i) + \sum_{i=0}^8 \delta(O_i) = 1.$$

Proof. Each exact slice D_k has density 2^{-k} . Hence

$$\delta(E_i) = \sum_{q \geq 0} 2^{-(2+2i+18q)} = \frac{2^{-(2+2i)}}{1 - 2^{-18}},$$

and

$$\delta(O_i) = \sum_{q \geq 0} 2^{-(1+2i+18q)} = \frac{2^{-(1+2i)}}{1 - 2^{-18}}.$$

Summing all normalized phases gives

$$\sum_{k \geq 1} 2^{-k} = 1.$$

$\square$

The normalized phases account for all one-step dyadic valuation mass of $\mathcal{O}$. This is exact local occupancy, not an asymptotic replacement for surjectivity.

13. FINITE WORDS AND DENSITY COST

For a finite valuation word $K = (k_1, \dots, k_j)$, define

$$S(K) = k_1 + \dots + k_j.$$

The density cost of the word is the product of the costs of its exact valuation steps.

Proposition 13.1 (Finite-word density cost). *A finite connected word K has density cost*

$$2^{-S(K)}.$$

At digit level $L \geq S(K) + 1$, its count is

$$2^{L-S(K)-1}.$$

Proof. A one-letter word with exponent k has cost 2^{-k} . Appending a valuation exponent k' selects one exact dyadic sub-slice of relative weight $2^{-k'}$. Multiplying these weights over the word gives

$$2^{-k_1} \dots 2^{-k_j} = 2^{-S(K)}.$$

The finite digit-level count follows by multiplying this relative density by 2^{L-1} . $\square$

14. SCALING EQUIVALENCE AND EXACT ROOTED OCCUPANCY

The preceding parts establish the required ingredients:

- local rails are exact, disjoint, and exhaustive;
- normalized phases account for total local density;
- word connection is deterministic;
- refinement is invariant;
- gaps propagate through scale;
- nontrivial cycles and repeat-hit masking are excluded.

Theorem 14.1 (No invariant scaling gap). *The rooted normalized construction has no invariant scaling gap.*

Proof. Assume a rooted occupancy gap occurs. Choose the first finite scale at which it appears. By the invariant gap propagation lemma, this gap has missing descendants at every higher scale. The gap therefore gives a persistent deficit in the invariant refinement geometry.

The local dyadic valuation rails already partition $\mathcal{O}$ exactly. The normalized E/O phases reorganize those rails and preserve their total finite-scale density. A persistent rooted deficit therefore requires one of three alternatives: an overlapping rooted hit, a duplicate count of the same value, or a broken phase connection.

Overlap and duplicate counting contradict rooted uniqueness together with the acyclicity theorem. A broken phase connection contradicts the mod 54 word-connection rule, which computes the next admissible side directly from the output residue. The remaining possibility is a local valuation gap, and the local rail partition excludes it.

Thus the assumed gap has no allowable source. The rooted normalized construction has no invariant scaling gap. $\square$

Theorem 14.2 (Exact finite-level rooted occupancy). *For every $L \geq 1$,*

$$\mathcal{R}_1 \cap \mathcal{O}_{<2^L} = \mathcal{O}_{<2^L}.$$

Proof. The rooted component is a subset of $\mathcal{O}$. The local rail partition gives the target occupancy at every finite digit level. The normalized phase densities match the target local density. The no invariant scaling gap theorem excludes a missing rooted value at any scale. Rooted uniqueness excludes surplus count from overlap or repeat hits. Therefore the rooted component occupies exactly the target odd set below every digit bound:

$$\mathcal{R}_1 \cap \mathcal{O}_{<2^L} = \mathcal{O}_{<2^L}.$$

$\square$

15. ROOTED SURJECTIVITY AND THE COLLATZ CONSEQUENCE

Theorem 15.1 (Rooted surjectivity). *The rooted inverse component equals the odd integers:*

$$\mathcal{R}_1 = \mathcal{O}.$$

Proof. Let $m \in \mathcal{O}$. Choose L with $m < 2^L$. Exact finite-level rooted occupancy gives

$$m \in \mathcal{R}_1 \cap \mathcal{O}_{<2^L}.$$

Thus $m \in \mathcal{R}_1$. Since every odd integer belongs to the rooted component, $\mathcal{R}_1 = \mathcal{O}$. $\square$

Corollary 15.2 (Collatz conjecture). *Every positive integer reaches 1 under the Collatz iteration.*

Proof. Every positive integer has the form

$$N = 2^a m,$$

with $m \in \mathcal{O}$. Dividing by powers of 2 brings N to m . Rooted surjectivity places m in the inverse component of 1, so the forward odd-to-odd trajectory from m reaches 1. The full trajectory then reaches the usual cycle

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

□

16. CONCLUSION

The proof establishes rooted surjectivity of the odd inverse Collatz system. Nontrivial cycles are excluded by the finite-word obstruction. The exact dyadic valuation rails partition the odd integers without gap or overlap. The normalized E/O phase system and mod 54 word-connection rule show how rooted segments combine. The refinement rule is invariant, so any misalignment propagates through every higher scale. Exact local occupancy, deterministic connection, and absence of repeat hits exclude persistent gaps. Therefore

$$\mathcal{R}_1 = \mathcal{O}.$$

Consequently, every positive integer reaches 1.

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